KAIZEN MATHS

Lesson starter

Binomial Expansion

Advanced Binomial Applications

Name _____

S1

Find the constant term in $(3x - x^{-1})^6$

S2

Use binomial expansion to approximate 1.02^6 (4 terms)

S3

In the expansion of $(3 - 2px)^9$, the coefficient of x^2 is equal to the coefficient of x^3. Find p.

S4

Find the constant term in $(4x - x^{-1})^6$

Show your working

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Show your working

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Exit ticket

Binomial Expansion

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Name _____

E1

Use binomial expansion to approximate 0.97^6 (4 terms)

E2

In the expansion of $(2-5px)^9$, the coefficient of x^2 is equal to the coefficient of x^3. Find p.

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Find the constant term in $(5x-3x^{-1})^6$

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Teacher answers & working

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S1

Find the constant term in $(3x - x^{-1})^6$

Answer:

-540

Working:

Method focus: use the power of x in the general term and set it equal to zero.

$T_{r+1} = {}^6C_r(3x)^{6-r}(-x^{-1})^r$

$T_{r+1} = {}^6C_r 3^{6-r}(-1)^r x^{6-2r}$

For constant term, $6 - 2r = 0$, so $r = 3$

constant $= {}^6C_3 3^3(-1)^3 = -540$

Final answer: -540

S2

Use binomial expansion to approximate 1.02^6 (4 terms)

Answer:

≈ 1.126160

Working:

Method focus: rewrite the number as 1 plus a small value and keep the requested number of terms.

$1.02^6 = (1 + 0.02)^6$

$(1 + x)^6 \approx 1 + 6x + {}^6C_2x^2 + {}^6C_3x^3$

$1 + 0.1200 + 0.006000 + 0.00016000 = 1.126160$

Actual value: 1.126162

Final answer: ≈ 1.126160

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S3

In the expansion of $(3-2px)^9$, the coefficient of x^2 is equal to the coefficient of x^3. Find p.

Answer:

$p = -\frac{9}{14}$

Working:

Method focus: find both coefficients using the general term, set them equal, then solve for p.

$T_{r+1} = {}^9C_r 3^{9-r}(-2px)^r$

Coefficient of x^2: ${}^9C_2 3^7 2^2 p^2$

Coefficient of x^3: $-{}^9C_3 3^6 2^3 p^3$

Set the coefficients equal and divide by $3^6 2^2 p^2$.

${}^9C_2 3 = -{}^9C_3 2p$

$p = -\frac{{}^9C_2 3}{{}^9C_3 2} = -\frac{9}{14}$

Final answer: $p = -\frac{9}{14}$

S4

Find the constant term in $(4x - x^{-1})^6$

Answer:

-1280

Working:

Method focus: use the power of x in the general term and set it equal to zero.

$T_{r+1} = {}^6C_r (4x)^{6-r}(-x^{-1})^r$

$T_{r+1} = {}^6C_r 4^{6-r}(-1)^r x^{6-2r}$

For constant term, $6 - 2r = 0$, so $r = 3$

constant $= {}^6C_3 4^3(-1)^3 = -1280$

Final answer: -1280

Teacher answers & working

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E1

Use binomial expansion to approximate 0.97^6 (4 terms)

Answer:

≈ 0.832960

Working:

Method focus: rewrite the number as 1 plus a small value and keep the requested number of terms.

$0.97^6 = (1 + -0.03)^6$

$(1+x)^6 \approx 1 + 6x + {}^6C_2x^2 + {}^6C_3x^3$

$1 + -0.1800 + 0.013500 + -0.00054000 = 0.832960$

Actual value: 0.832972

Final answer: ≈ 0.832960

E2

In the expansion of $(2 - 5px)^9$, the coefficient of x^2 is equal to the coefficient of x^3. Find p.

Answer:

$p = -\frac{6}{35}$

Working:

Method focus: find both coefficients using the general term, set them equal, then solve for p.

$T_{r+1} = {}^9C_r 2^{9-r}(-5px)^r$

Coefficient of x^2: ${}^9C_2 2^7 5^2 p^2$

Coefficient of x^3: $-{}^9C_3 2^6 5^3 p^3$

Set the coefficients equal and divide by $2^6 5^2 p^2$.

${}^9C_2 2 = -{}^9C_3 5p$

$p = -\frac{{}^9C_2 2}{{}^9C_3 5} = -\frac{6}{35}$

Final answer: $p = -\frac{6}{35}$

Teacher answers & working

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E3

Find the constant term in $(5x - 3x^{-1})^6$

Answer:

-67500

Working:

Method focus: use the power of x in the general term and set it equal to zero.

$T_{r+1} = {}^6C_r(5x)^{6-r}(-3x^{-1})^r$

$T_{r+1} = {}^6C_r 5^{6-r}(-3)^r x^{6-2r}$

For constant term, $6 - 2r = 0$, so $r = 3$

constant $= {}^6C_3 5^3(-3)^3 = -67500$

Final answer: -67500